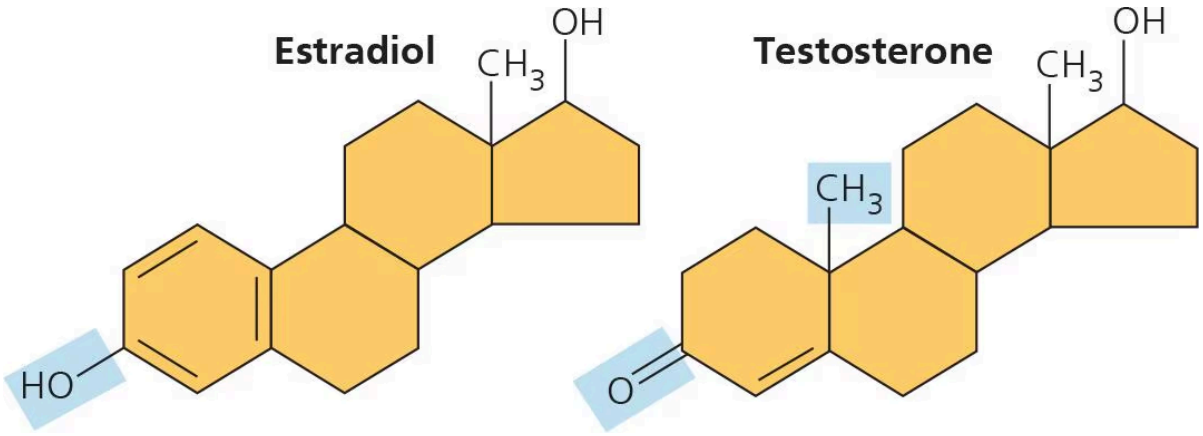
# 4.3: A few chemical groups are key to molecular function

The distinctive properties of an organic molecule depend not only on the arrangement of its mostly carbon skeleton but also on the various chemical groups attached to that skeleton. These groups may participate in chemical reactions or may contribute to function indirectly by their effects on molecular shape; they help give each molecule its unique properties.

## The Chemical Groups Most Important in the Processes of Life

Consider the differences between estradiol (a type of estrogen) and testosterone. These compounds are female and male sex hormones, respectively, in humans and other vertebrates. Both are steroids, organic molecules with a common carbon skeleton in the form of four fused rings. They differ only in the chemical groups attached to the rings (shown here in abbreviated form, where each corner represents a carbon and its attached hydrogens); the distinctions in molecular architecture are shaded in blue:




In estradiol, one of four fused rings is aromatic, containing three double bonds, while in testosterone, this ring contains only one double bond. Estradiol has an alcohol (O H) where testosterone has a ketone, and testosterone has a methyl (C H 3) group that is not present in estradiol.

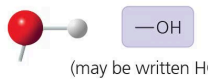
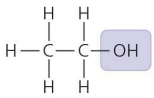
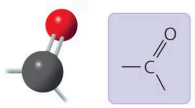
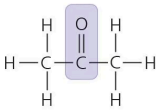
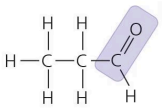
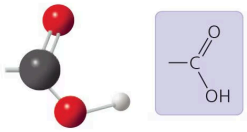
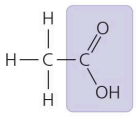
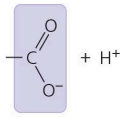
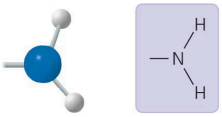
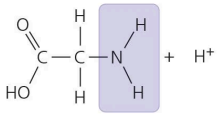
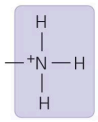
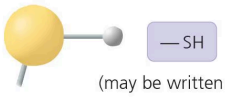
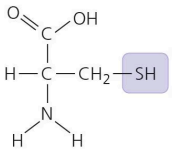
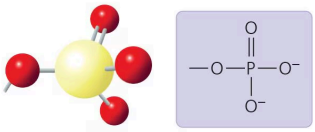
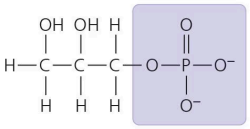
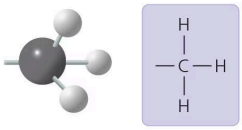
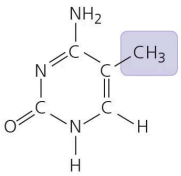
The different actions of these two molecules on many targets throughout the body are the basis of sexual characteristics, producing the contrasting features of male and female vertebrates. In this case, the chemical groups are important because they affect molecular shape, contributing to function.

In other cases, chemical groups are directly involved in chemical reactions; such groups are known as **functional groups**. Each has certain properties, such as shape and charge, that cause it to participate in chemical reactions in a characteristic way.

The seven chemical groups most important in biological processes are the **hydroxyl group**, **carbonyl group**, **carboxyl group**, **amino group**, **sulfhydryl group**, **phosphate group**, and **methyl group**. The first six groups can be chemically reactive; of these six, all except the sulfhydryl group are also hydrophilic and thus increase the solubility of organic compounds in water. The methyl group is not reactive, but instead often serves as a recognizable tag on biological molecules. Study **Figure 4.9** to become familiar with these biologically important chemical groups. Note in the Examples column of the figure that the carboxyl group and the amino group are ionized at normal cellular pH.

**Figure 4.9**

Some biologically important chemical groups.

| Chemical Group | Group Properties and Compound Name | Examples |
|---|---|---|
| **Hydroxyl group** (—OH)   | Is polar due to electronegative oxygen. Forms hydrogen bonds with water, helping dissolve compounds such as sugars.<br><br>Compound name: **Alcohol** (specific name usually ends in *-ol*) |  **Ethanol**, the alcohol present in alcoholic beverages |
| **Carbonyl group** (>C=O)  | Sugars with ketone groups are called ketoses; those with aldehydes are called aldoses.<br><br>Compound name: **Ketone** (carbonyl group is within a carbon skeleton) or **aldehyde** (carbonyl group is at the end of a carbon skeleton) |  **Acetone**, the simplest ketone  **Propanal**, an aldehyde |
| **Carboxyl group** (—COOH)  | Acts as an acid (can donate $H^+$) because the covalent bond between oxygen and hydrogen is so polar.<br><br>Compound name: **Carboxylic acid**, or **organic acid** |  **Acetic acid**, which gives vinegar its sour taste  Ionized form of —COOH (carboxylate ion), found in cells |
| **Amino group** (—$NH_2$)  | Acts as a base; can pick up an $H^+$ from the surrounding solution (water, in living organisms).<br><br>Compound name: **Amine** |  **Glycine**, an amino acid (note its carboxyl group)  Ionized form of —$NH_2$, found in cells |
| **Sulfhydryl group** (—SH)   | Two —SH groups can react, forming a "cross-link" that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair; in hair salons, "permanent" treatments break cross-links, then re-form them while the hair is in the desired shape.<br><br>Compound name: **Thiol** |  **Cysteine**, a sulfur-containing amino acid |
| **Phosphate group** (—$OPO_3^{2-}$)  | Contributes negative charge (1– when positioned inside a chain of phosphates; 2– when at the end). When attached, confers on a molecule the ability to react with water, releasing energy.<br><br>Compound name: **Organic phosphate** |  **Glycerol phosphate**, which takes part in many important chemical reactions in cells |
| **Methyl group** (—$CH_3$)  | Affects the expression of genes when bonded to DNA or to proteins that bind to DNA. Affects the shape and function of male and female sex hormones.<br><br>Compound name: **Methylated compound** |  **5-Methylcytosine**: Cytosine, a component of DNA, has been modified by addition of a methyl group. |

The columns are chemical group, group properties and compound names, and examples. Row entries are as follows.

Chemical group: Hydroxyl group (single bond, O H). The group is polar and can form hydrogen bonds. Compound name: Alcohol. Hydroxyl groups have the structure single bond, O H. The group is polar and can form hydrogen bonds (for example, with water, helping dissolve compounds such as sugars). Sugars have

many hydroxyl groups. Compounds that contain an hydroxyl group are also called alcohols. An example is ethanol, the alcohol in alcoholic beverages.

Chemical group: Carbonyl group, which has the structure C O, where a carbon and oxygen are double bonded. Compound name: ketone or aldehyde. Carbonyl groups have the structure C O, where a carbon and oxygen are double bonded. Carbonyl groups at the end of a carbon chain are aldehydes (these also have a hydrogen on the carbonyl carbon). Carbonyl groups in the middle of a carbon chain are ketones. Propanol, with three carbons, is an aldehyde. Acetone, also with three carbons, is the simplest ketone. A sugar containing an aldehyde group is called an aldose, and a sugar containing a ketone group is called a ketose.

Chemical group: Carboxyl group (structure C O O H). Acts as acid (can donate H plus) because the covalent bond between oxygen and hydrogen is so polar. Compound name: Carboxylic acid or organic acid. Carboxyl groups have the structure single bond, C O O H. The carbon has a double bond to one oxygen and a single bond to the O H. Carboxyl groups act as acids, ionizing to form carboxylate ion, single bond, C O O minus and H plus. A carboxylate ion is the same as a carboxyl group, but has a negative charge on the oxygen that lost a hydrogen. Compounds containing the carboxyl group are known as carboxylic acids or organic acids. An example is acetic acid, a two-carbon acid that gives vinegar its sour taste. Ionized form of single bond, C O O H (carboxylate ion), is found in cells.

Chemical group: Sulfhydryl group (single bond, S H). Compound name: Thiol. Sulfhydryl groups have the structure single bond, S H. Two sulfhydryl groups can react, forming a disulfide cross-link of the form single bond, S, single bond, S, single bond, that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair. In hair salons, these crosslinks are broken and re-formed during a permanent treatment. Compounds containing sulfhydryl groups are known as thiols. An example is the amino acid cysteine, which contains a sulfhydryl group at the end of its side-chain.

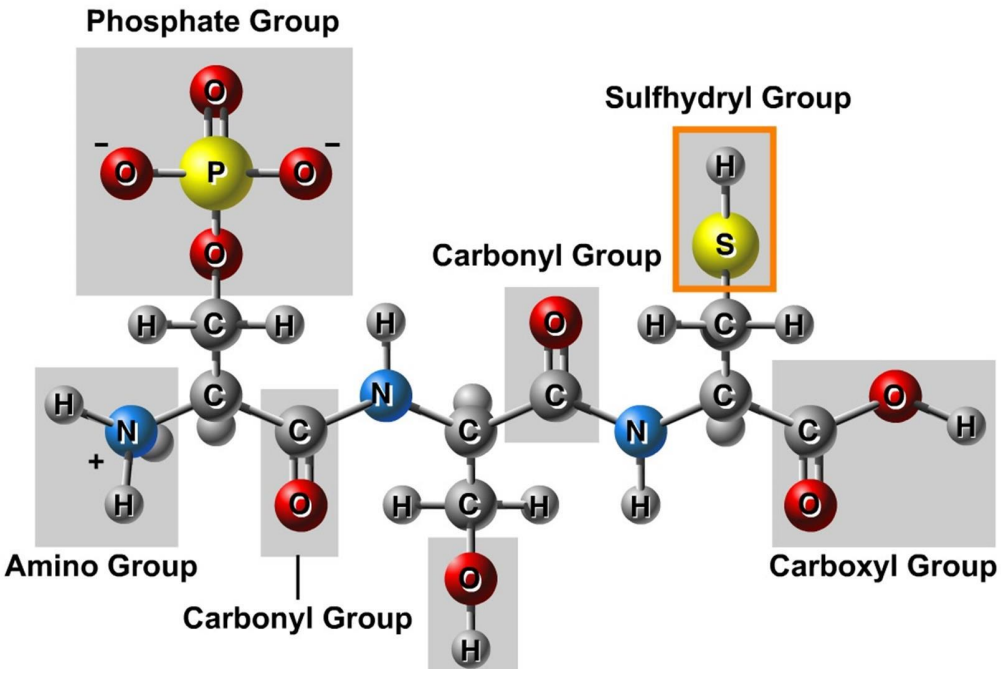
Chemical group: Phosphate group (single bond, O P O 3 superscript minus 2). Contributes negative charge (1 minus when position inside a change of phosphates; 2 minus when at the end). Compound name: Organic phosphate. Phosphate groups consist of a phosphorus atom surrounded by four oxygen atoms. They can bond to a carbon or other atom through one or two of their four oxygen atoms. The structure can be written as single bond, O P O 3 superscript minus 2. They have a charge of minus 2 when positioned at the end of a chain, or minus 1 when in the middle of a chain. When an attached phosphate group is hydrolyzed (broken off) by water, energy is released. Carbon compounds containing phosphate are called organic phosphates. An example is glycerol phosphate, a three-carbon compound with a phosphate on carbon 1, and hydroxy groups on carbons 2 and 3.

Chemical group: Methyl group (single bond, C H 3). Affects the expression of genes when on DNA or on proteins bound to DNA. Compound name: Methylated compound. Methyl groups have the structure single bond, C H 3, and are ubiquitous in organic chemistry. When present in D N A or proteins bound to D N A,

they can affect gene expression. The presence of methyl groups also affects the shape and function of male and female sex hormones. Compounds containing methyl groups can be called methylated compounds. An example is 5-methyl cytosine, the methylated form of the D N A base cytosine.

**Animation**

Functional Groups

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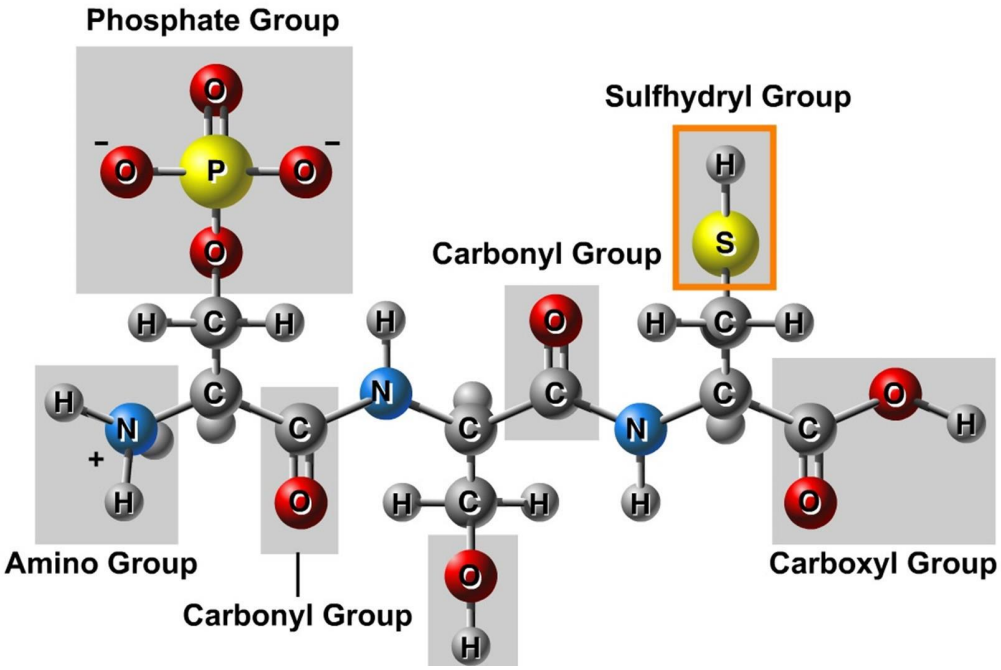

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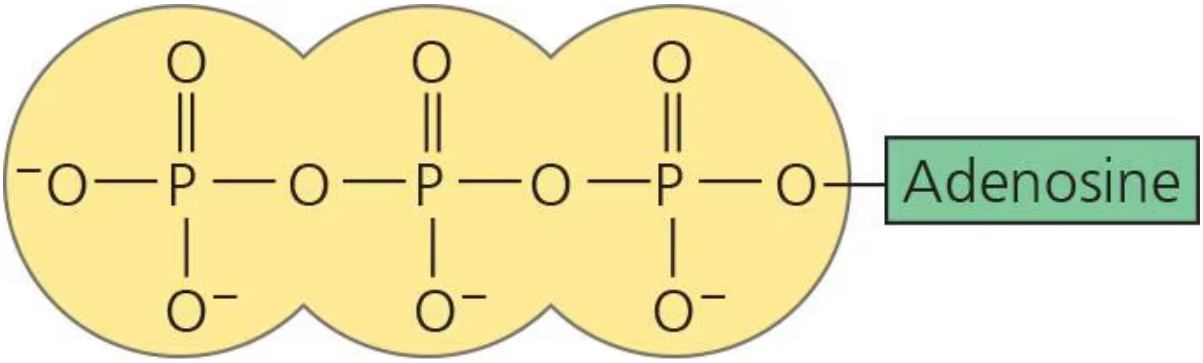
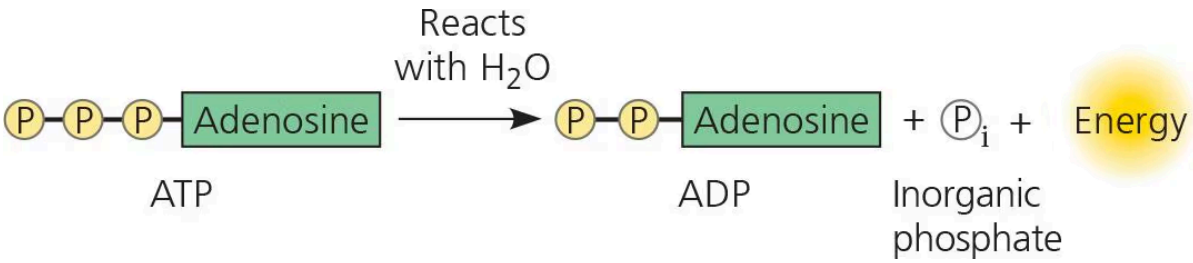
## ATP: An Important Source of Energy for Cellular Processes

The “Phosphate group” row in Figure 4.9 shows a simple example of an organic phosphate molecule. A more complicated organic phosphate, **adenosine triphosphate**, or **ATP**, is worth mentioning here because its function in the cell is so important. ATP consists of an organic molecule called adenosine attached to a string of three phosphate groups:




Adenosine attached to a tail of three phosphate groups. Each phosphate group consists of a central P that forms 5 bonds: a double bond to one O and three single covalent bonds to three other oxygen atoms. Oxygen atoms forming only single covalent bonds in this molecule are shown with negative charges.

Where three phosphates are present in series, as in ATP, one phosphate may be split off as a result of a reaction with water. This inorganic phosphate ion, HOPO32−, is often abbreviated ℗i in this book, and a phosphate group in an organic molecule is often written as ℗. Having lost one phosphate, ATP becomes adenosine *di*phosphate, or ADP. Although ATP is sometimes said to store energy, it is more accurate to think of it as storing the potential to react with water or other molecules. Overall, the process releases energy that can be used by the cell. You’ll learn more about this in Concept 8.3.




The reactant A T P (an adenosine attached to 3 P’s representing phosphate groups) reacts with water, producing P sub i (inorganic phosphate), A D P (adenosine attached to 2 P’s) and energy.

## The Chemical Elements of Life: *A Review*

Living matter, as you have learned, consists mainly of carbon, oxygen, hydrogen, and nitrogen, with smaller amounts of sulfur and phosphorus. These elements all form strong covalent bonds, an essential characteristic in the architecture of complex organic molecules. Of all these elements, carbon is the virtuoso of the covalent bond. The versatility of carbon makes possible the great diversity of organic molecules, each with particular properties that emerge from the unique arrangement of its mostly carbon skeleton and the chemical groups attached to that skeleton. This variation at the molecular level provides the foundation for the rich biological diversity found on our planet.